

Andre Arcaina

 andrearcaina  andrearcaina  andrearcaina.ca

Education

Toronto Metropolitan University (TMU, formerly Ryerson University)

Sept 2022 – Present

Bachelor of Science (Honours), Computer Science (Co-op)

Toronto, ON

Coursework: Data Structures, Algorithms, Databases, Computer Networks, Operating Systems, Software Engineering, Object Oriented Programming, Distributed Systems, Big Data Systems

Work Experience

Environment and Climate Change Canada (ECCC)

June 2025 – Present

Software Engineer Intern

North York, ON

- Identified lack of automated testing in Jenkins CI and built a Go-based end-to-end testing framework to validate component data flow across multiple test cases, reducing bug triage time in satellite data pipelines.
- Led modernization of mission-critical Java services by migrating 8 components to Java 17 and Jakarta EE 10, improving maintainability and supporting national weather data delivery systems used in maritime and Coast Guard operations.
- Eliminated decades of technical debt by rebuilding a legacy VB6 application from the ground up in Python/PyQt with a PostgreSQL backend, securely processing datasets exceeding 50,000 rows.
- Led the documentation and maintainability of a legacy C++ and Java system, authoring foundational architecture and UML diagrams to accelerate future feature development for a 3-person team.
- Refactored 3 legacy C++ services by removing deprecated exception patterns, modernizing error handling, and improving code clarity, reducing technical debt and improving long-term maintainability across production systems.

Undergraduate Science Society of TMU (USSTM)

Jan 2025 – Apr 2026

Backend Engineer

Toronto, ON

- Architected a campus vault backend for the undergraduate science student body by engineering a contract-first Go RESTful API across 32+ endpoints, leveraging SQLc for type safety and OpenAPI for strict schema enforcement.
- Led backend reliability initiatives by developing 37+ unit and integration tests, achieving 85%+ code coverage and preventing regressions in core business logic.
- Built AWS S3 image storage with presigned URLs to support secure, horizontally scalable before and after item image uploads.
- Developed asynchronous email processing using Redis-backed Asynq workers and AWS SES for reliable notification delivery.
- Collaborated with the frontend team to deliver scalable platform features aligned with organizational requirements.

DataKinetics

May 2024 – Aug 2024

Software Engineer Intern

Ottawa, ON

- Eliminated manual data inspection for the R&D team by developing a Spring Boot RESTful middleware API and web dashboard that automated COBOL-to-JSON parsing using an external Java library, documented end-to-end in Confluence.
- Presented the finalized tool to 12 R&D stakeholders, successfully demonstrating the new automated data-inspection workflow.
- Designed and executed 72 comprehensive Postman test cases for a core mainframe REST API, rigorously validating JSON responses, edge cases, and data integrity across 3 IBM Db2 databases.

Projects

Distributed Stock Trading Platform | Go, gRPC, Docker, Kubernetes, NATS, PostgreSQL, Redis

GitHub

- Spearheaded architecture of a distributed backend for a fintech paper trading platform composed of 8+ microservices, handling 1,600+ RPS across 5,000 concurrent users during load testing with a 0% failure rate (P95: 42ms, Avg: 7ms).
- Led infrastructure deployment using Kubernetes and Helm, integrating system observability with Prometheus, Loki and Grafana.
- Drove a 98% improvement in market data processing latency (800ms to 10ms) through Redis caching and request coalescing.

Concurrent Codebase Analysis CLI & SDK | Go, Cobra, Bubble Tea, Concurrency

GitHub | Package

- Engineered a multi-threaded static analysis CLI capable of concurrently scanning 46,000+ file monorepos in under 0.7 seconds, achieving >600% CPU utilization on multi-core CPUs.

Skills

Languages: Go, Python, Java, C++, Rust, JavaScript, TypeScript, NoSQL, SQL, GraphQL

Frameworks: FastAPI, Django, Spring Boot, Node.js, React.js, Next.js, TailwindCSS, gRPC

Tools: GCP, AWS, Docker, Kubernetes (Minikube), NATS, Postgres, Redis, Linux, Git, Bash, CI/CD, Jenkins, GitHub Actions